

Zirui Zhao | Curriculum Vitae

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Education

Computer Science

Ph.D., Singapore

Advisor: Prof. Lee Wee Sun

CAP:5.0/5.0

School of Computing, National University of Singapore

2020 - Now

Automation(Qian Class)

B.Eng., Xi'an, China

Advisor: Prof. Pengju Ren

Thesis: Robustify Route-based Visual Place Recognition by Bridging Kalman Filter with CNN Representation

EECS & Qian Xuesen College, Xi'an Jiaotong University

2016 - 2020

GPA:3.9/4.3

Visiting

Safe AI Lab

Summer intern, Pittsburgh, USA

Advisor: Prof. Ding Zhao

Carnegie Mellon University

2019 Summer

School of Computing

Visiting student, Singapore

Advisor: Prof. Soo Yuen Jien

National University of Singapore

2018 Summer

Publications & Manuscripts

[2]: R. Chen, W. Wang, **Z. Zhao**, D. Zhao, "Active Learning for Risk-Sensitive Inverse Reinforcement Learning," submitted to ICRA 2020, preprinted in arXiv.

[1]: **Z. Zhao**, Y. Mao, Y. Ding, P. Ren and N. Zheng, "Visual-Based Semantic SLAM with Landmarks for Large-Scale Outdoor Environment," 2019 2nd China Symposium on Cognitive Computing and Hybrid Intelligence (CCHI), Xi'an, China, 2019, pp. 149-154.

Selected Research Projects

Safe AI Lab

Active Risk-sensitive Inverse Reinforcement Learning

Risk-sensitive inverse reinforcement learning provides an general model to capture how human assess the distribution of a stochastic outcome when the true distribution is unknown (ambiguous). This work enables an RS-IRL learner to actively query expert demonstrations for faster risk envelope approximation.

CMU

2019 Summer

Cognitive Architecture Lab

Visual Semantic SLAM

This work combines the semantic labels with pointclouds generated from ORB-SLAM by using bayesian update rules. Besides, the landmarks in the real world is attached in the map, associated with the pointclouds by using fuzzy mathematics.

XJTU

Nov. 2018 - Jun. 2019

Cognitive Architecture Lab

Multi-robot Cooperative navigation

This project established a multi-robot navigation and exploration system, which consist of UAVs and UGVs.

XJTU

Apr. 2018 - Mar. 2019

Demo video link: <https://www.youtube.com/watch?v=wSB4hyW9rWc&t=5s>

Selected Scholarships & Honors

2020: NUS Research Scholarship

2020: XJTU Outstanding Graduate Honor

2019/2018/2017: XJTU Outstanding Student Honor

2018: XJTU 2nd Prize of 1989 Mechanical Engineering Alumni Scholarship for Qian Xue Sen Program

2017: XJTU Siyuan Merit Scholarship

Interests & Skills

Interests: Robotics, Task and Motion Planning, Robot Learning, Cognitive Robotics, Human-Robot-Interaction.

Language: C++, Python, MATLAB, $\LaTeX$

Tools: TensorFlow, PyTorch, Keras, Scikit-Learn, Numpy, Jupyter, OpenCV, ROS.

Software: Carla, GitLab, GitHub.